Mini TCP

ENGINEERING RELIABILITY OVER UNRELIABLE BEST-EFFORT NETWORKS

Name	PANCHAL HARSHIL
Enrollment No.	240021704100028
Roll No.	G1_310

CORE ARCHITECTURE & OBJECTIVES

- **Protocol Foundation** Stateful, reliable transport protocol designed and implemented atop UDP.
- **Core Mechanisms** Connection management, state transitions, error detection, and loss recovery.

VERIFICATION & SIMULATION

- **Synthetic Loss Injection** Python test harness (30% drop rate)
- **Network Simulation** Cisco Packet Tracer simulation

Needs & Requirements: Transport Reliability

BRIDGING THE GAP BETWEEN BEST-EFFORT IP ROUTING AND MISSION-CRITICAL BYTE-STREAM DELIVERY

BEST-EFFORT TRANSPORT (IP/UDP)

IP and UDP offer zero delivery guarantees, resulting in inherent network layer instabilities for enterprise workloads.

ZERO GUARANTEES
Routinely Dropped Packets

Buffer overflows and network congestion cause unannounced packet loss with no native recovery mechanism.

UNRELIABLE PATHS
Packet Reordering & Corruption

Multi-path routing causes out-of-order delivery, while bit errors degrade payload integrity without application-layer safety.

CRITICAL GAP:

Enterprise workloads strictly require **100% byte-stream fidelity** to operate correctly.

THE MINI TCP SOLUTION

A custom transport protocol layer introducing deterministic reliability, order, and integrity over IP/UDP.

EXACT-ORDER RECONSTRUCTION
Byte-Level Sequencing

Tracks every individual payload byte sequentially to guarantee precise, perfectly ordered reconstruction at the destination.

AUTOMATED RECOVERY
Positive ACK with Retransmission (PAR)

Automatically detects dropped or missing packets using deterministic timers, triggering swift retransmission.

PAYLOAD VALIDATION
End-to-End Integrity Verification

Rigorously validates both header correctness and payload accuracy before passing clean data to the application layer.

PROTOCOL_STACK: [IP / UDP] -> [MINI-TCP ENGINE] -> [APPLICATION]

STATUS: GUARANTEED_BYTE_STREAM_DELIVERY

Protocol Details: Mini TCP Header & Framing

LIGHTWEIGHT BINARY FRAMING ENCAPSULATED INSIDE UDP DATAGRAMS (PORT 9999)

01. Binary Header Frame Format (12 Bytes Total) Bit 0 ---------- Bit 31

SEQUENCE NUMBER [HDR1]
Tracks Initial Sequence Number (ISN) & exact byte-offset
32 BITS (BYTES 0-3)

ACKNOWLEDGMENT NUMBER [HDR2]
Indicates the next expected byte sequence from peer
32 BITS (BYTES 4-7)

CONTROL FLAGS 8 BITS
SYN | ACK | DATA | FIN state mask

PAYLOAD LENGTH 16 BITS
Defines chunk size (50 bytes max)

CHECKSUM 16 BITS
Validates header + payload integrity

WIRESHARK PACKET CAPTURE — LOOPBACK INTERFACE (LO0) — UDP PORT 9999

```
[UDP Datagram Payload] Src Port: 54102 -> Dst Port: 9999 | Length: 34 bytes
0000 00 00 00 01 00 00 00 00 04 00 16 A2 48 65 6C 6C |............Hell|
0010 6F 20 57 6F 72 6C 64 21 |o World! |
```

▲ Header Breakdown: Seq=0x00000001 (1) | Ack=0x00000000 (0) | Flags=0x04 (DATA) | Len=0x0016 (22 B) | Csum=0xA248

Resolutions & Core Reliability Mechanisms

DETERMINISTIC TIMEOUT (RTO), LOSS RECOVERY, AND CHECKSUM INTEGRITY UNDER 30% LOSS

~400 ms

Fixed RTO

Utilizes a dedicated non-blocking socket timer for precise frame tracking.

- If no valid ACK arrives within the window, a client timeout triggers an immediate retransmit interrupt.

Deduplication

Dual-Failure Recovery

- **Scenario A (Data Loss):** Standard retransmission on timeout.
- **Scenario B (ACK Loss):** Client resends data. Server detects duplicate byte range, discards redundant payload, and re-transmits ACK.

Frame Integrity

Checksum Validation

- Computed directly prior to packet transmission and verified upon payload read.
- Mismatched frames are silently dropped at the receiver level, forcing clean timeout recovery without corrupting buffers.

Winding Up: Mini TCP vs. TCP (Pros & Cons)

ARCHITECTURAL EVALUATION, TRADE-OFF MATRIX, AND VERIFICATION CONCLUSIONS

DIMENSION	MINI TCP	STANDARD TCP (RFC 9293)
Flow Control	**Stop-and-Wait ARQ** Restricted to 1 segment in flight. Causes severe throughput degradation on high-latency links.	**Sliding Window** Dynamic window sizing allows multiple in-flight segments, maximizing bandwidth-delay product fill.
Timeout Calculation	**Static ~400 ms RTO** Fixed timer fails to adapt to variable network jitter, frequently triggering premature or stalled recovery.	**Adaptive RTO (Karn/Partridge)** Continuously measures RTT variance to calculate dynamic retransmission timeouts accurately.
Congestion Control	**No Exponential Backoff** Lacks backoff algorithms during loss, risking router queue collapse under extreme retry loops.	**Congestion Algorithms** Implements Slow Start, Congestion Avoidance, Fast Retransmit, and exponential binary backoff.
State Machine	**Simplified 3-State Model** Omits TIME_WAIT and complex teardown phases, leading to potential teardown hangs or orphan packets.	**Full 11-State FSM** Includes TIME_WAIT (2MSL) and structured graceful closure to ensure reliable connection termination.

Future Scope & Protocol Roadmap

NEXT-PHASE ENHANCEMENTS: SLIDING WINDOWS, ADAPTIVE RTT, AND FULL CONNECTION TEARDOWN

01 // PIPELINING

Sliding Windows

Implement Go-Back-N or Selective Repeat sliding window protocols to allow multiple unacknowledged packets in flight simultaneously, maximizing link capacity.

02 // ADAPTIVE TIMERS

Dynamic RTT & Backoff

Introduce Dynamic RTT Estimation combined with Exponential Backoff to auto-tune timeout clocks dynamically under changing network jitter.

03 // TEARDOWN

TIME_WAIT State

Integrate a full TIME_WAIT teardown state to automatically recover dropped teardown packets and guarantee clean, unambiguous socket closures.

THANK

YOU